python

Programming 1

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Lecture #10

- downloading webpages
- JSON serialization

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Downloading webpages

```
4  import urllib.request
5
6  def main():
7      response = urllib.request.urlopen("http://python.org")
8      raw = response.read()
9      print(type(raw))    # bytes
10     # print(raw)
11
12     html = raw.decode("utf-8")
13     print(type(html))    # str
14     # print(html)
```

Since it's a common operation, let's put it in our `pylab.py` module in a function called `get_page()` . We want to use it like this:

```
3  from pylab import get_page
4
5  url = 'http://python.org'
6
7  def main():
8      print(get_page(url))
```

Downloading a URL (it can be an HTML, an image, etc.).
The downloaded object will appear in the filesystem:

```
4 import urllib.request
5
6 url = "http://www.nhdfl.org/uploads/NHB%20photos/HBOP2.jpg"
7
8 def main():
9     urllib.request.urlretrieve(url, "/tmp/forest.jpg")
```

Or, you can also use an external program to download something, e.g. `wget` :

```
3  import os
4
5  url = 'http://wallpapers.leovacity.be/images/Forrest_wallpaper.jpg'
6  to = '/tmp/forrest.jpg'
7
8  def main():
9      cmd = 'wget {url} -O {output}'.format(url=url, output=to)
10     print(cmd)
11     os.system(cmd)
```

calling an external program in another process

The **requests** module

The Python standard library has the following modules for downloading URLs:

- urllib (https://docs.python.org/3/library/urllib.html)
- urllib.request (https://docs.python.org/3/library/urllib.request.html)
- ...

However, if you want to do something more complex, they are not that easy to use :(

There is a very useful module called **requests** (link). It 's not part of the standard library, you need to install it separately. But it's much simpler to use than the two modules mentioned above…

Presentation from its author.

Installation:

```
sudo apt-get install python3-pip
sudo pip3 install pip -U
sudo pip3 install requests -U
```

Run it once (if the package manager pip is not yet installed).

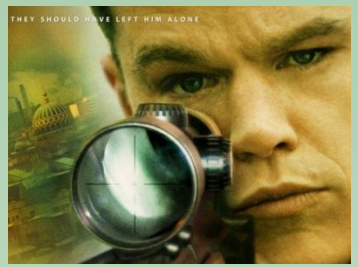
Update the pip package manager to the newset version.

install/update requests

How to use the **requests** module

```
>>> import requests
>>>
>>> url = "http://python.org"
>>> r = requests.get(url)
>>> print(r.text)
```

JSON serialization



- http://json.org/
- http://en.wikipedia.org/wiki/Json

With the help of the **json** module you can serialize Python objects residing in the memory to JSON format.

Serialization

Converting a data structure or object to a format that can be stored in a file (or can be transmitted through network), and later either in the same, or in a different computing environment it can be „resurrected".

data structure, object $\xrightarrow{\text{serialization}}$ file, bit stream

data structure, object $\xleftarrow{\text{de-serialization}}$ file, bit stream

Formats: XML, JSON, YAML, stb.

JSON can be often used as an alternative of XML

- also text-based, easily readable for humans
- also hierarchical
- can also be used for communication between applications
- simpler than XML
- not as verbose, shorter
- can be written / read faster

XML:

```
<Person>
   <FirstName>Homer</FirstName>
   <LastName>Simpson</LastName>
   <Relatives>
      <Relative>Grandpa</Relative>
      <Relative>Marge</Relative>
      <Relative>The Boy</Relative>
      <Relative>Lisa</Relative>
   </Relatives>
</Person>
```

JSON:

```
{
   "firstName": "Homer",
   "lastName": "Simpson",
   "relatives": [ "Grandpa", "Marge", "The Boy", "Lisa" ]
}
```

XML:

```
<persons>
 <person>
  <name>Ford Prefect</name>
  <gender>male</gender>
 </person>
 <person>
  <name>Arthur Dent</name>
  <gender>male</gender>
 </person>
 <person>
  <name>Tricia McMillan</name>
  <gender>female</gender>
 </person>
</persons>
```

JSON:

```
[
  {
    "name": "Ford Prefect",
    "gender": "male"
  },
  {
    "name": "Arthur Dent",
    "gender": "male"
  },
  {
    "name": "Tricia McMillan",
    "gender": "female"
  }
]
```

XML:

```
<settings>
    <path>/home/luke/Dropbox/Public</path>
    <user_id>123456</user_id>
    <auto_sync>True</auto_sync>
</settings>
```

(source: Dropbox Publicus)

JSON:

```
{
   "path": "/home/luke/Dropbox/Public",
   "user_id": 123456,
   "auto_sync": true
}
```

JSON Syntax Rules

JSON syntax is a subset of the JavaScript object notation syntax.

- Data is in name/value pairs
- Data is separated by comma
- Curly brackets holds objects
- Square brackets holds arrays

JSON Values

JSON values can be:

- A number (integer or floating point)
- A string (in double quotes)
- A Boolean (true or false)
- An array (in square brackets)
- An object (in curly brackets)
- null

loading a JSON **file**

input file (person.json)

```
1  {
2      "last": "Doe",
3      "first": "John",
4      "age": 39,
5      "sex": "M",
6      "registered": true,
7      "salary": 70000
8  }
```

pretty print
(data structures are printed in a nicer way)

json module
(writing / reading json files)

```
3   from pprint import pprint
4   import json
5
6
7   def read_file():
8       f = open('person.json', 'r')
9       d = json.load(f)
10      f.close()
11
12      print(type(d))
13      pprint(d, indent=4)
```

```
{     'age': 39,
      'first':  'John',
      'last':  'Doe',
      'registered': True,
      'salary': 70000,
      'sex':  'M'}
```

output (dictionary)

returned value: Python data structure
(here in the example: dictionary)

loading a JSON **string**

```
7   def read_string():
8       s = """{
9       "path": "/home/luke/Dropbox/Public",
10      "user_id": 123456,
11      "auto_sync": true
12  }"""
13
14      d = json.loads(s)
15
16      print(d)
```

(see settings.py)

loads
(**s** stands for string)

Summary

The *json* module takes a string (either from a file or from a normal string) and **builds a Python object** (e.g. list, dictionary). That is, we don't need to parse the string manually, the module performs this task for us!

Exercises

Asking my own IP address

Link: https://arato.inf.unideb.hu/szathmary.laszlo/pmwiki/index.php?n=EnPy3.20120920g

Downloading an image from Reddit

Link: https://arato.inf.unideb.hu/szathmary.laszlo/pmwiki/index.php?n=EnPy3.20121126a

Writing data structures to **file** in JSON format

(see write_person.py)

```
3   import json
4
5   def write_file():
6       person = {
7           'last': 'Doe',
8           'first': 'John',
9           'age': 39,
10          'sex': 'M',
11          'registered': True,
12          'salary': 70000,
13      }
14
15      f = open('employee.json', 'w')
16      json.dump(person, f)
17      f.close()
```

Python dictionary

json.dump(data_structure, file_handler)

```
print(json.dumps(person))
```

it creates a JSON string from the data structure and returns this JSON string

```
json.dump(person, f, indent=4)
```

pretty print

```
json.dump(person, f,
    indent=4, sort_keys=True)
```

pretty print, ordered by keys

Exercises

homework

1. [20120815k] prison break